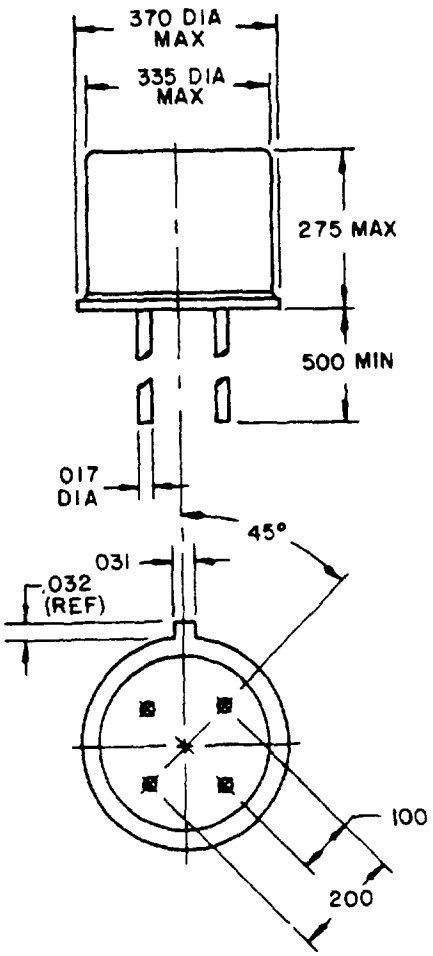
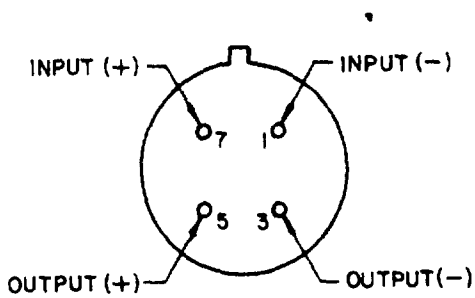
MILITARY SPECIFICATION SHEET

Ⓓ RELAY, SOLID STATE, HERMETICALLY SEALED, ISOLATED, ±50 MA, ±40 V, AC OR DC, ANALOG SIGNAL SWITCHING, SPST (N.O.)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-R-28750





CIRCUIT
(BOTTOM VIEW)

INCHES	MM
.017	.43
.031	.79
.032	.81
.100	2.54
.200	5.08
.275	6.99
.335	8.51
.370	9.40
.500	12.70

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm).
4. Terminal numbers shown above are for reference only

FIGURE 1. Outline drawing and dimensions.

Ⓓ denotes changes

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REQUIREMENTS

INPUT DATA

Input voltage range 4.0 to 7 V dc. 1/

Rated turn-on voltage 5.0 V dc. 1/

Rated turn-off voltage 1.0 V dc maximum. 1/

Input current 22 mA dc maximum at rated voltage.

Turn-on time 10 microseconds maximum at rated voltage.

Turn-off time 15 microseconds maximum from rated voltage.

Bias voltage Not applicable.

OUTPUT DATA

Rated output current See figure 2.

Rated output voltage ±40 V, ac or dc.

Output voltage drop .5 V dc maximum at rated voltage.

Output leakage current 100 microamperes maximum at rated voltage.

Crosstalk Not applicable.

Transient voltage Not applicable.

Electric system spike Not applicable.

Overload 0.01 joule surge, 1 percent duty cycle.

DC offset voltage ±10 millivolts

Waveform distortion Not applicable.

ELECTRICAL DATA

Dielectric withstanding voltage 1,000 V ac (P-P), 60 Hz.

Insulation resistance 100 megohms.

(D) Isolation 10 picofarads tested at 1 kHz.

Power dissipation 140 milliwatts maximum at rated voltage

ENVIRONMENTAL DATA

Temperature (operating and storage) -55°C to +125°C.

(D) Shock (specified pulse) MIL-STD-202, method 213, test condition F (1500 G's).

(D) Vibration MIL-STD-202, method 204, test condition H, except peak value shall be 100 G's (10 to 2,000 Hz).

Electromagnetic interference Not applicable

Resistance to soldering heat Applicable

Salt spray (corrosion) In accordance with MIL-STD-750, method 1041.

Moisture resistance Not applicable.

1/ Input must be a step function. Rise or fall time, as applicable, not to exceed 100 microseconds

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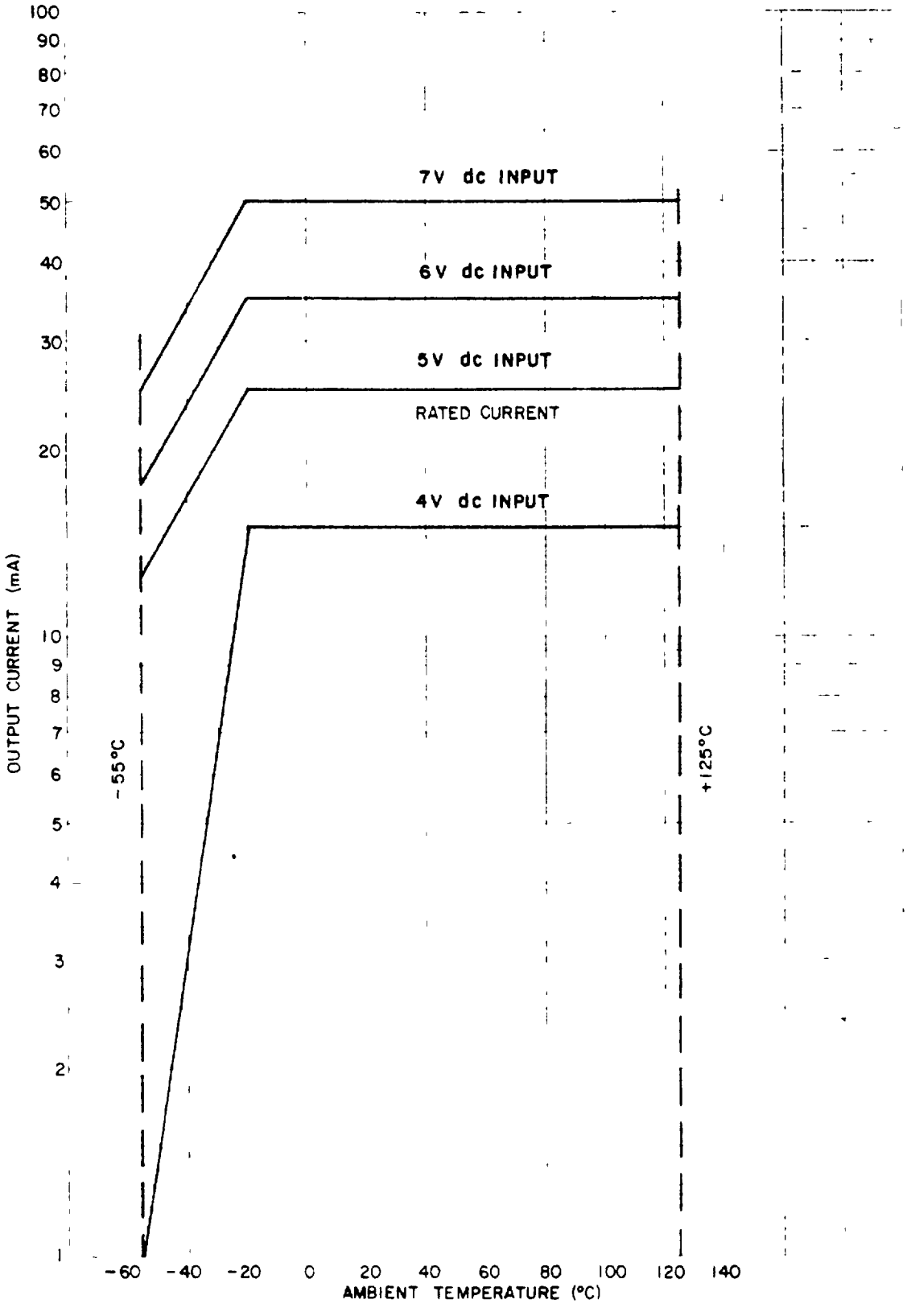


FIGURE 2. Output current vs input control voltage and ambient temperature.

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PHYSICAL DATA.

Terminal strength (MIL-STD-202, method 211)

Pull test Test condition A, 1 pound pull.

Bend test Test condition C, 1/2 pound load.

Twist test Test condition D.

Terminal solderability Applicable.

Dimensions and configuration. See figure 1.

Weight 5 grams (.18 ounce) typical.

Seal. Hermetic.

LIFE TEST REQUIREMENTS.

Life test Applicable.

Minimum marking. Military part number, functional diagram and terminal identification, date code, and manufacturer's name or source code.

(D) Part number M28750/5-001W, Y, or X to indicate the applicable screening level as specified in MIL-R-28750.

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, AT, ME
Navy - AS, MC, SH
Air Force - 19

Agent
DLA - ES

Preparing activity
Navy - EC

(Project 5945-0682-1)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-28750/15D	Relay, Solid State, Hermetic

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION (*Mark one*)
	☐ VENDOR
b ADDRESS (*Street, City, State, ZIP Code*)	☐ USER
	☐ MANUFACTURER
	☐ OTHER (*Specify*) _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER (*Last, First, MI*) – Optional	b WORK TELEPHONE NUMBER (*Include Area Code*) – Optional
c MAILING ADDRESS (*Street, City, State, ZIP Code*) – Optional	8 DATE OF SUBMISSION (*YYMMDD*)

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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